

ML-Cloud

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Setup

Documentation:

[h !\[\]\(b96b3a660a85c4a0498f921ce823c64a_img.jpg\) //portal.mlcloud.uni-tuebingen.de/](https://portal.mlcloud.uni-tuebingen.de/)

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(Bernhard, Katrin, Zehao)

Option 1 - add keypair on ML-Cloud

Generate new private, public key pair on the mlcloud.

Add the public key into the path/to/home/.ssh/authorized_keys folder.

Since all the nodes share the same file system they will recognize this new public key.

You can then ssh to the gpu node by using the private key you generated on the ML cloud.

1. Generate key: `ssh-keygen -t ecdsa -b 521`
2. add generated public key to path/to/home/.ssh/authorized_keys via adding it to a new line
3. start gpu node e.g.: `salloc --partition=gpu-2080ti-interactive --gres=gpu:1 --no-shell`
4. login to gpu node: `ssh -i /path/to/mlcloud_private_key [name of machine e.g.: slurm-bm-16]`
-> -i option might not be necessary!
5. (start tmux/ screen or do whatever you want)
6. logout after you are done: `logout`

Option 2 - Forwarding ssh agent

(Last test if this is working is already a while ago - Katrin)

1. **Before** connecting to the cluster run `ssh-add`
2. Connect to the cluster with `ssh -A <clustername>`
3. Start job - works with interactive jobs and sbatch jobs (probably also other commands)
4. Connect to node with `ssh <nodename>`

2. start interactive session

1. `salloc --partition=gpu-2080ti-interactive --gres=gpu:1 --no-shell`
2. `ssh -i /path/to/mlcloud_private_key [name of machine e.g.: slurm-bm-16]`



Some useful tools + commands

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1. **Tutorial**

Please check it here: <https://github.com/baumgach/tue-slurm-helloworld>



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Pip: slurm-gpustats



Under AVG Wiki management Personnel/Contracts Organisation Administration

There are a total of 608 gpus [up]

4 gtx1080ti gpus

64 v100 gpus

540 rtx2080ti gpus

There are a total of 584 gpus [accessible]

4 gtx1080ti gpus

56 v100 gpus

524 rtx2080ti gpus

Usage by user:

```
ipetrov55 [total: 1 (interactive: 0 )] rtx2080ti: 1
sramrakhiani43 [total: 1 (interactive: 0 )] rtx2080ti: 1
jbader40 [total: 1 (interactive: 0 )] rtx2080ti: 1
mgorecki90 [total: 1 (interactive: 1 )] rtx2080ti: 1
szabel09 [total: 1 (interactive: 0 )] rtx2080ti: 1
reschenhagen26 [total: 1 (interactive: 0 )] rtx2080ti: 1
jschmidt14 [total: 1 (interactive: 0 )] rtx2080ti: 1
pmayilvahanan31 [total: 1 (interactive: 0 )] rtx2080ti: 1
cbaumgartner [total: 1 (interactive: 1 )] rtx2080ti: 1
abhattacharyya22 [total: 1 (interactive: 0 )] v100: 1
pghosh58 [total: 2 (interactive: 0 )] rtx2080ti: 2
tklein16 [total: 2 (interactive: 0 )] rtx2080ti: 2
opress [total: 2 (interactive: 0 )] rtx2080ti: 2
mtangemann [total: 2 (interactive: 0 )] rtx2080ti: 2
fmohamed65 [total: 5 (interactive: 0 )] rtx2080ti: 5
krenz73 [total: 5 (interactive: 0 )] rtx2080ti: 5
mweis [total: 7 (interactive: 0 )] rtx2080ti: 7
nkister74 [total: 8 (interactive: 0 )] rtx2080ti: 4, v100: 4
lsalewski11 [total: 8 (interactive: 0 )] v100: 8
mkuemmerer31 [total: 9 (interactive: 0 )] rtx2080ti: 9
sun22 [total: 9 (interactive: 0 )] rtx2080ti: 4, v100: 5
svolz67 [total: 12 (interactive: 0 )] rtx2080ti: 12
bjaeger25 [total: 16 (interactive: 0 )] rtx2080ti: 8, v100: 8
ysharma [total: 17 (interactive: 0 )] rtx2080ti: 5, v100: 12
jcforno31 [total: 18 (interactive: 0 )] rtx2080ti: 18
tmenge30 [total: 22 (interactive: 0 )] rtx2080ti: 22
mbini75 [total: 24 (interactive: 0 )] rtx2080ti: 24
zstojanovski18 [total: 32 (interactive: 0 )] rtx2080ti: 32
```

There are 377 gpus available:

rtx2080ti: 355 available

v100: 18 available

gtx1080ti: 4 available

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3. Priority

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```
sshare -a
```

4. Delete all jobs with same job name

```
SQUEUE_FORMAT2="jobid:10,partition:4,state:8,username:7,starttime:20,timeuse
d:11,timelimit:2,nodelist:8,tres-per-node:6,name:130" squeue | grep
<username> | grep <job-name> | cut -d ' ' -f1 | xargs qdel
```

Attention: This command directly cancels the jobs! If you first want to display the PIDs execute the command without "qdel".

5. Delete files

Create a function in your ~/.bashrc file

```
# combined delete for mlcloud
```

```
rmrf(){
```

```
    for arg; do
```

```
        find "${arg}" -type f ! -size 0c | parallel -X --progress
```

```
truncate -s0
```

```
        rm -rf "${arg}"
```

```
    done
```

```
}
```

Then you can delete files with `rmrf abd*`.

5. Fast copy to another file system

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```
ls --indicator-style=none /path/to/qb/dataset | xargs -n1 -P4 -I% rsync -Pa %  
/path/to/lustre/dataset/
```

The / at the end is important. It will copy the folders in the root level of your folder in parallel to the second path (here lustre).

-P4 means 4 parallel processes.

Your working directory needs to be /path/to/qb/dataset

Submitting jobs

Might be useful but haven't tested yet: <https://github.com/albanie/yaspi>

1. Job arrays

Used by Bozidar?, Katja?, Axel?

Haiwen: I used sh files adapted from this repository: <https://github.com/y0ast/slurm-for-ml>

TODO

2. Custom Python file

Used by Katrin

TODO

Personal workflows and experiences



1. ssh to login node

2. if not already running: start interactive session

-> since i am using a python script to start jobs and this should not be executed on the login node i start my jobs from the interactive session.

-> This also has the advantage that i have one GPU allocated and can do minor debugging on the ML-Cloud.

-> Another advantage: if they re-start the login nodes i still have my tmux sessions :)

3. ssh to interactive session

4. tmux

5. submit jobs

Copy large dataset to region 2:

This sends your folder (unencrypted!) to the other node.

Was much faster for me than rclone when I tried to copy data from a VM to region 2 SLURM.

sender node tmux:

```
export IP=IP_of_target_node
```

finds a free port:

```
export PORT=$( ssh username@$IP -i /path/to/private_key '/usr/sbin/ss -Htuan' | awk '{gsub(/.*:/,"",$5); print $5"\n"$5} END{for(i=2220;i<=2230;i++) print i}' | sort | uniq -u | shuf | head -1 )
```

```
echo $PORT
```

receiver node tmux:

replace port with outputted port above

```
nc -l -p PORT | tar -C /path/to/target_folder -xf -
```

sender node:

replace IP with IP of receiver node

```
tar -c /path/to/source/folder | pv -b | nc $IP $PORT
```



Remote SSH Plugin in Visual Studio Code

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Troubleshooting on ML-Cloud

Author: Daniel

The Remote SSH plugin in VS code has several benefits when editing or running code on a remote machine or the ML-Cloud. The integration in VS Code is handy and the experience for editing, searching, saving is significantly faster and more stable (once a connection is established). However, sometimes a connection cannot be established to the ML-Cloud or AVG VMs and I found these steps helpful:

1. **Adding a symlink:** VS code typically downloads a `.vscode-server` folder onto your home directory. This folder can easily exceed 1-2GB, which is a problem on the ML-Cloud and AVG VMs. Therefore, it's recommended to copy the folder to a different directory and adding a symlink, e.g.

```
mv /home/geiger/$USER/.vscode-server /mnt/qb/work/geiger/$USER/.vscode-server
ln -s /mnt/qb/work/geiger/$USER/.vscode-server /home/geiger/$USER/.vscode-server
```

2. **Adjusting the settings:** A certain setting in VS code is sometimes not properly synced between the local and remote VS code installations. As a results, a connection can sometimes not be established. Simply check the box in Settings->Extensions->Remote - SSH->Remote: Download Extensions Locally or add this line to your JSON config: `"remote.downloadExtensionsLocally": true`. Afterwards, you need to refresh your VS Code window (Ctrl+Shift+P -> Developer: Reload Window) and your connection is hopefully running again. Using a different log-in node can sometimes also be beneficial.

Stalking students

```
sacct -u gwbXXX --starttime 2025-02-02 --
```

```
format=User,JobID,Jobname%50,partition,state,time,start,end,elapsed,MaxVMSize,nnodes,nodelist,ReqTRES%50
```



